DS-UA 202: Responsible Data Science
Course Project Draft Report

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Project Overview

ADS we propose to analyze: The Algorithmic Decision-Support System (ADS) we want to analyze for this project is a predictive customer churn model designed for a music streaming service called Playlist Pro. We are going to be auditing a PyTorch-based Multi-Layer-Perceptron (MLP) neural network that processes user demographics (age, location) and behavioral data (listening hours, customer inquiries, etc.) to predict whether or not a customer will cancel their subscription. The system's output is meant to support a direct business decision: determining which "at-risk" users will be targeted and offered financial retention discounts.

Why we chose this ADS (in relation to the course): We chose this system because it shows us how standard business algorithms, like predicting churn to allocate marketing budgets, can inadvertently create equity and fairness issues for consumers. Because the dataset includes demographic features like age and location, it provides an ideal testbed to audit whether the model's errors disproportionately exclude certain protected or vulnerable groups from receiving valuable financial discounts. This directly aligns with our course's focus on evaluating the real-world, human impact of algorithmic interventions rather than just measuring raw predictive accuracy.

Sources:

- **Kaggle Competition:** https://www.kaggle.com/competitions/streaming-subscription-churn-mode leaderboard
- **Implementation Solution:** https://www.kaggle.com/code/ethanjychen/streaming-churn-rate

1 Background

1.1 Purpose and Stated Goals

The purpose of this ADS is to predict which Playlist Pro subscribers are likely to cancel their subscription (churn) so the company can proactively offer them financial retention discounts. The stated goal is to accurately identify at-risk customers before they leave, allowing the marketing team to allocate retention budgets to the right people rather than blanketing all users with offers.

1.2 Trade-offs and Multiple Goals

The main trade-off is between catching as many churners as possible (recall) versus not wasting discounts on people who would have stayed anyway (precision). A model that flags everyone as

at-risk will catch all churners but costs the company unnecessary discounts; a model that is too conservative will miss real churners, and those users lose out on offers that might have kept them. This becomes an equity issue if the model's misses are concentrated in specific demographic groups, as those groups are systematically excluded from a financial benefit. There is also a tension between optimizing for overall accuracy versus ensuring the model performs fairly across age groups and locations.

2 Input and Output

2.1 Data Source and Collection

The data comes from the Kaggle competition "Streaming Subscription Churn Model." It is a synthetic dataset simulating what a real music streaming platform would collect from its subscription system, activity logs, and support tickets. Each row represents one customer.

2.2 Data Profiling and Distributions

The input features are:

- `customer_id`: Integer, unique identifier.
- `age`: Integer, subscriber age, roughly 18–70+, uniformly distributed with extreme frequencies on opposite ends
- `location`: Categorical, geographic region of the subscriber, uniformly distributed
- `subscription_type`: Categorical, e.g., Basic/Standard/Premium, uniformly distributed.
- `payment_plan`: Categorical, e.g., monthly/yearly, uniformly distributed.
- `weekly_hours`: Float, total or average listening time, right-skewed with power users at the high end,
- `customer_service_inquiries`: Integer, number of support contacts, uniformly distributed across Low/Medium/High.

Note: because the dataset is synthetic, there are no missing values whatsoever.

Other Inputs: The dataset also includes 13 other behavioral and transactional features (such as `num_subscription_pauses`, `payment_method`, and `signup_date`). However, our profiling focuses on the features above as they demonstrated the highest correlation with churn and serve as the foundation for our fairness subgroups.

In terms of correlations, `customer_service_inquiries` tends to correlate positively with churn (see Figure 2) while `weekly_hours` correlates negatively (See Figure 1. The correlation between age and these behavioral features matters for our audit: if age predicts listening behavior, the model may use behavioral features as proxies for age even if age were removed.

2.3 System Output Interpretation

The output is a binary class label: churned (1) or not churned (0). Under the hood, the PyTorch MLP outputs a sigmoid probability between 0 and 1, thresholded at 0.5. Customers predicted as churned get placed on a retention target list and are offered discounts. Customers predicted as not churned receive no intervention. Therefore, the model's output directly determines who gets a financial benefit.

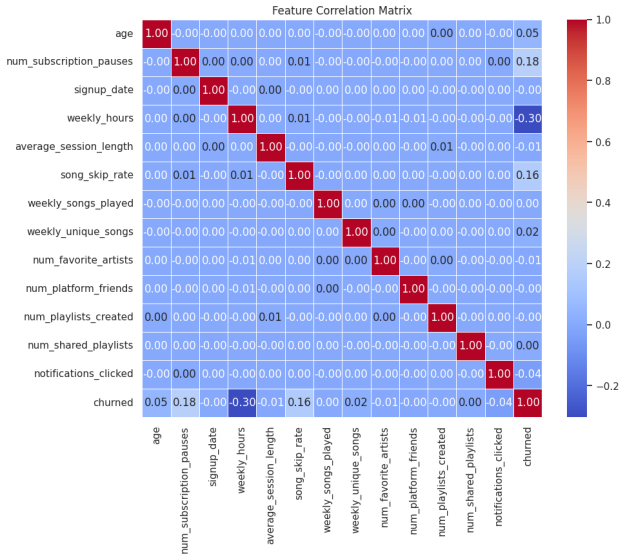


Figure 1: Feature Correlation Matrix

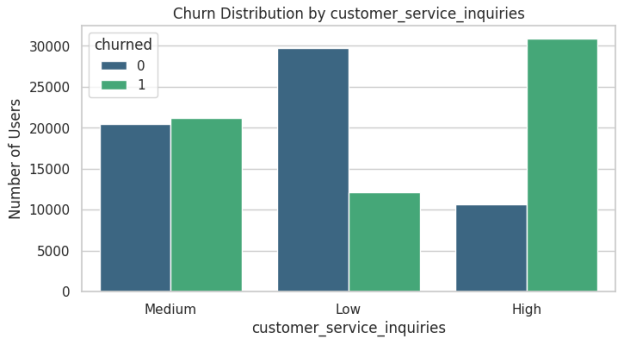


Figure 2: Churn Distribution by customer_service_inquiries

3 Implementation and Validation

3.1 Data Cleaning and Pre-processing

The solution notebook by ethanjychen encodes categorical variables (`location`, `subscription_type`, `payment_plan`) using one-hot or label encoding, normalizes numerical features (`age`, `listening_hours`, `customer_service_inquiries`) via standardization, drops `customer_id` since it has no predictive value, and splits the data into training and test sets.

3.2 Implementation details

The model is a Multi-Layer Perceptron (MLP) built in PyTorch. It is a feedforward neural network with an input layer, one or more hidden layers using ReLU activations, and a sigmoid output layer for binary classification. It is trained with binary cross-entropy loss and an Adam optimizer over multiple epochs. At inference time, each customer's features are passed through the network and the sigmoid output is thresholded at 0.5 to produce a churn/no-churn prediction.

3.3 Validation Methods

The original solution notebook submitted by the author did not explicitly calculate comprehensive validation metrics, relying solely on Kaggle's public leaderboard where it achieved an accuracy of roughly 84.67%. To establish a rigorous baseline for our audit, we implemented an evaluation step on a 20% held-out validation set.

We calculated the model's aggregate accuracy, precision, recall, F1-score, and AUC-ROC. **The model achieved an accuracy of 84.0%, Precision of 83.29%, Recall of 86.13%, F-1 Score of 84.69%, and AUC-ROC of 93.18%.** While these aggregate metrics confirm the model predicts churn reasonably well overall, they mask the underlying distribution of errors. They do not tell us whether the model performs equally well across demographic subgroups—which is exactly what our fairness audit will examine.

4 Detailed Audit Plan

Our audit will focus on whether the model's errors are distributed equitably across demographic groups defined by age and location. The plan has five steps:

1. Define subgroups by binning age into brackets (e.g., 18–25, 26–35, 36–50, 51+) and using the existing location categories, including intersectional groups.

2. Compute precision, recall, false positive rate, and false negative rate separately for each subgroup, with special attention to false negatives since those represent churners who are denied retention discounts.

3. Evaluate formal fairness metrics including demographic parity, equalized odds, and predictive parity across groups.

4. Use SHAP or permutation importance to assess how much the model relies on age and location directly versus using behavioral features as demographic proxies.

5. Simulate the downstream dollar impact by calculating how retention discounts would be distributed across subgroups under the model's predictions.

Based on our findings, we will recommend threshold adjustments, feature removal, or calibration strategies to improve fairness without unacceptably hurting accuracy.

5 Preliminary Performance Analysis & Early Takeaways

We have verified that the data is accessible and successfully run the baseline model code. Our early look at the data and existing solutions suggests that the model's strong aggregate accuracy (84–87%) likely masks uneven error rates across demographic groups. Specifically, `listening_hours` and `customer_service_inquiries`—the two most predictive behavioral features—appear to have distributions that vary by age bracket.

If the model has learned decision boundaries calibrated on the majority behavioral profile, it may systematically misclassify users whose engagement patterns differ. Since false negatives mean a churner does not get a retention discount, this could create a pattern where certain age groups or locations are less likely to receive a financial benefit not because of their actual churn risk, but because of how the model encodes their behavior. This reinforces why our audit needs to go beyond aggregate metrics and examine exactly who the model gets wrong.

Auditing a Streaming Subscription Churn Model

RESPONSIBLE DATA SCIENCE
FINAL PROJECT

Terry Han & Trem Ampeloquio

Group 87

Background

Overview

Welcome to the DSC 783 Final Project.

For this project, you are a Business Analyst working for a Music Streaming Service called PlaylistPro.

At PlaylistPro, one of company's top priorities is to **reduce customer churn (stop customers from cancelling their subscriptions).**

To reduce churn, PlaylistPro plans to reduce the cost of the music streaming service for customers that are likely to cancel their subscription. But, the company currently does not know which customers are likely or unlikely to cancel their subscription.

Thus, PlaylistPro would like you to build a Supervised Classification Model to predict if a customer will churn based on their subscription information and listening habits. To help you build a model, the company's Operations team created a dataset of 125,000 customers to train your model on. More details are below:

CLICK HERE TO VIEW THE KAGGLE COMPETITION

PURPOSE

- **The System**: PyTorch MLP Neural Network.
- **The Task**: Predict subscriber churn for PlaylistPro based on demographics and listening habits.
- **The Intervention**: Offer financial retention discounts to "at-risk" users.

CONCERNS

- **High Precision** (Too conservative) → Miss churners, unfarily exclude them from discounts
- **High Recall** (Too liberal) → Flag everyone; wastes company's marketing budget

Input & Output

INPUT

- SYNTHETIC DATASET SIMULATING STREAMING PLATFORM USER DATA, 1 ROW = 1 USER

- NOTABLE INPUTS FROM KAGGLE COMPETITION "STREAMING SUBSCRIPTION CHURN MODEL."
  - CUSTOMER_ID (INT)
  - AGE (INT)
  - SONG_SKIP_RATE (FLOAT)
  - AVERAGE_SESSION_LENGTH (FLOAT)
  - SUBSCRIPTION_TYPE (STR)
  - PAYMENT_PLAN (STR)
  - WEEKLY_HOURS (INT)
  - CUSTOMER_SERVICE_INQUIRIES (STR)

- *CUSTOMER_SERVICE_INQUIRIES* TENDS TO CORRELATE POSITIVELY WITH CHURN WHILE *WEEKLY_HOURS* CORRELATES NEGATIVELY

OUTPUT

- BINARY CLASS LABEL:

LABELLED CHURNED (1) → DISCOUNTS
LABELLED NOT CHURNED (0) → NO DISCOUNTS

- UNDER THE HOOD, THE PYTORCH MLP OUTPUTS A SIGMOID PROBABILITY BETWEEN 0 AND 1, THRESHOLDED AT 0.5.

Detailed Analysis Plan

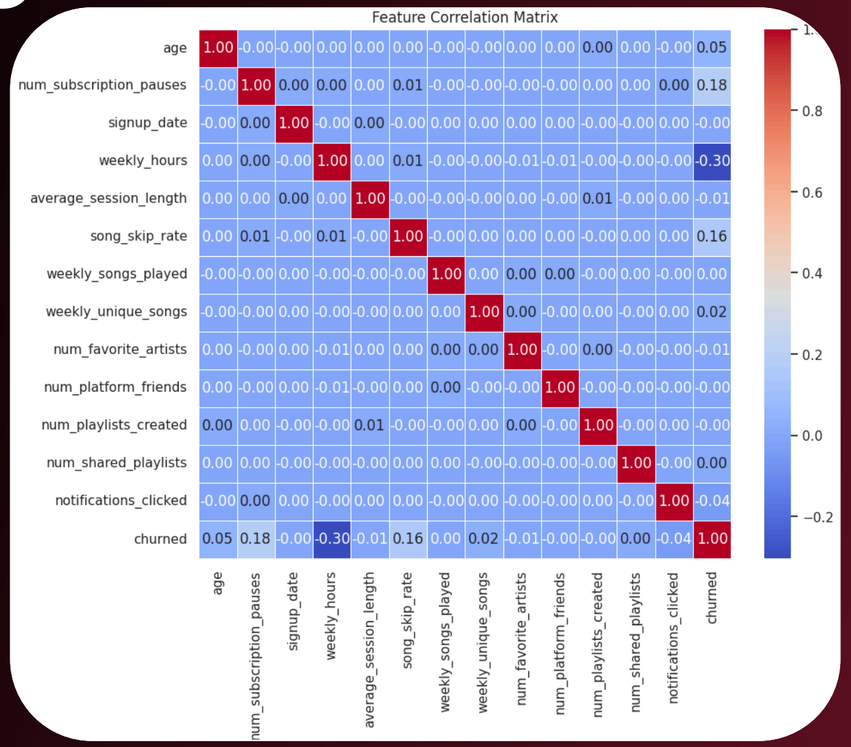
1. DEFINE AGE SUBGROUPS
2. COMPUTE PRECSION, RECALL, FPR, FNR*
3. EVALUATE DEMOGRAPHIC PARITY, EQUALIZED ODDS, PREDICTIVE PARITY
4. SHAP FOR FEATURE CONTRIBUTION
5. COMPUTE DOWNSTREAM DOLLAR IMPACT

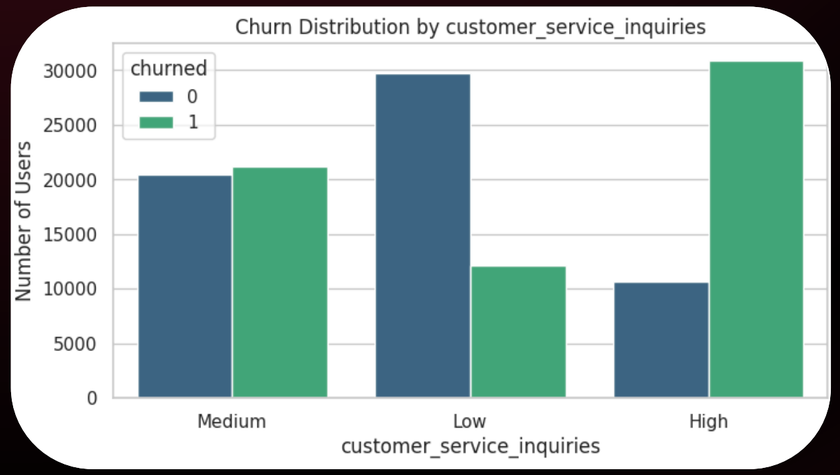
Based on findings we'll recommend threshold adjustments, feature removal, or calibration strategies to improve fairness without unacceptably hurting accuracy.

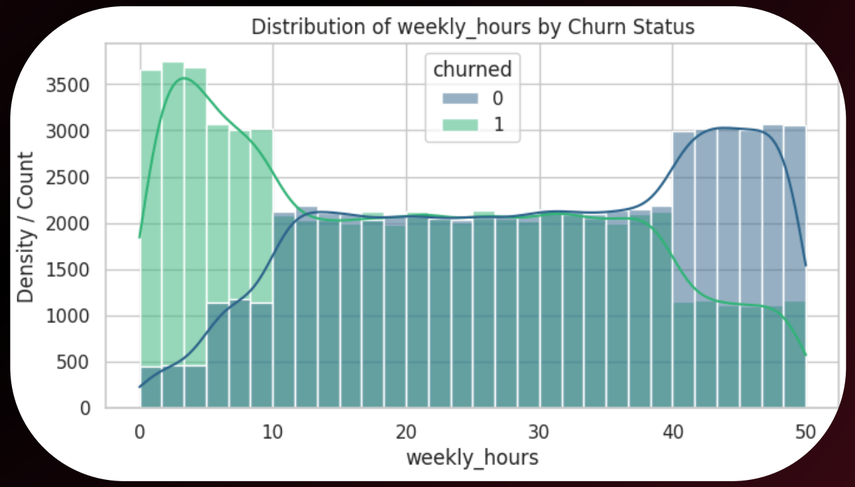
* special attention to FNR since those represent churners who are denied retention discounts

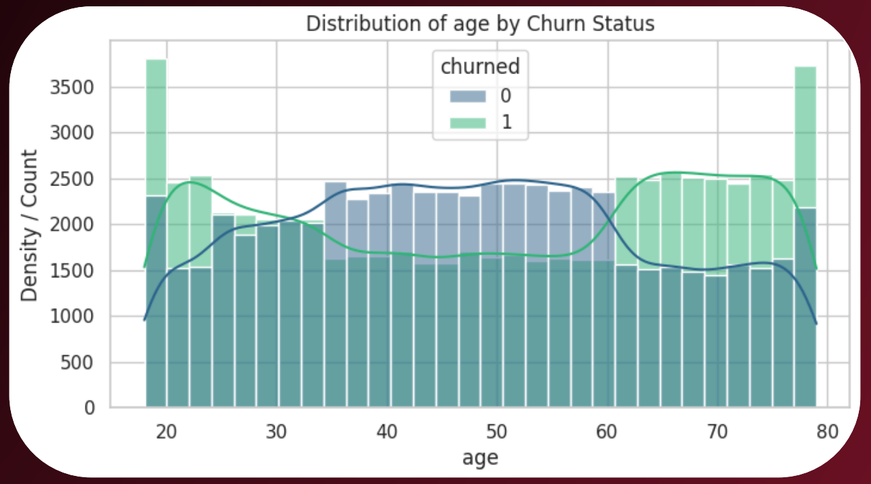
Early Takeaways

- MODEL'S STRONG AGGREGATE ACCURACY (~84–87%) LIKELY MASKS UNEVEN ERROR RATES ACROSS DEMOGRAPHIC GROUPS
- BEHAVIOR DRIVES THE MODEL
  - BEHAVIOR AS A PROXY TO DISCRIMINATE BASED ON AGE
- IF THE MODEL HAS A HIGHER FALSE NEGATIVE RATE FOR SPECIFIC AGE GROUPS, THOSE USERS ARE BEING SYSTEMATICALLY EXCLUDED FROM RECEIVING FINANCIAL RETENTION DISCOUNTS









Thank you

Questions?